

**C MAINE SHELF BUOY 44032, ME, USA**WMO: **997170**Lat: **43.715N**Lon: **69.355W**Elev: **0**StdP: **101.33**Time Zone: **-5.00 (NAE)**Period: **06-18**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -11.9      | -9.7 | N/A                           | N/A | N/A  | N/A | N/A | N/A  | 16.8                  | -0.5 | 15.8 | -2.0 | 9.0                   | 310  | 0.683 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 8             | 2.5                    | 20.8            | N/A  | 20.2 | N/A  | 19.5 | N/A  | N/A                 | N/A  | N/A | N/A  | N/A | N/A  | 3.9                  | 210  |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| N/A                             | N/A | N/A  | N/A | N/A | N/A  | N/A | N/A | N/A  | N/A           | N/A  | N/A  | N/A  | N/A  | N/A  | N/A               |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (d)                        | (e)   | (f)                | (g) | (h)                                                | (i)   | (j)        | (k)   | (l)        | (m)   | (n)        | (o)   | (p)  | (q) |
| (4) 15.0          | 13.7 | 11.7 |                            |       |                    |     |                                                    |       |            |       |            |       |            |       |      |     |
| (5)               |      |      | DB                         | -14.2 | 22.6               | 1.3 | 1.3                                                | -15.1 | 23.5       | -15.9 | 24.3       | -16.6 | 25.0       | -17.6 | 26.0 |     |
|                   |      |      | WB                         | N/A   | N/A                | N/A | N/A                                                | N/A   | N/A        | N/A   | N/A        | N/A   | N/A        | N/A   | N/A  |     |

**Monthly Climatic Design Conditions**

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 8.4    | -0.8  | -1.2  | 1.4   | 5.7   | 9.5   | 13.4  | 17.2  | 17.5  | 15.5  | 11.5  | 7.2   | 2.8   |
|                                                                        | DBStd       | 7.18   | 4.91  | 4.01  | 3.28  | 1.96  | 2.01  | 1.90  | 1.82  | 1.52  | 1.78  | 2.43  | 3.47  | 4.34  |
|                                                                        | HDD10.0     | 1410   | 335   | 314   | 265   | 128   | 36    | 0     | 0     | 0     | 0     | 14    | 93    | 224   |
|                                                                        | HDD18.3     | 3656   | 593   | 548   | 523   | 378   | 275   | 147   | 44    | 34    | 86    | 211   | 334   | 482   |
|                                                                        | CDD10.0     | 813    | 0     | 0     | 0     | 0     | 18    | 103   | 222   | 234   | 165   | 62    | 9     | 0     |
|                                                                        | CDD18.3     | 17     | 0     | 0     | 0     | 0     | 0     | 0     | 7     | 9     | 1     | 0     | 0     | 0     |
|                                                                        | CDH23.3     | 1      | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDH26.7     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 6.0    | 8.2   | 7.9   | 7.4   | 5.6   | 4.7   | 4.0   | 3.8   | 3.8   | 4.7   | 6.4   | 7.2   | 7.9   |
| (15) Precipitation                                                     | PrecAvg     | 1240   | 100   | 89    | 120   | 106   | 96    | 94    | 90    | 77    | 110   | 121   | 123   | 115   |
|                                                                        | PrecMax     | 1784   | 181   | 191   | 211   | 219   | 218   | 224   | 179   | 197   | 242   | 383   | 216   | 192   |
|                                                                        | PrecMin     | 871    | 15    | 50    | 23    | 12    | 26    | 34    | 39    | 9     | 42    | 24    | 55    | 43    |
|                                                                        | PrecStd     | 221    | 43    | 30    | 51    | 51    | 46    | 54    | 43    | 48    | 57    | 76    | 39    | 35    |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 10.2  | 7.3   | 7.8   | 11.6  | 15.5  | 18.9  | 22.0  | 21.9  | 20.5  | 16.9  | 14.0  | 11.8  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 2%          | DB     | 8.8   | 6.2   | 6.8   | 9.8   | 14.3  | 17.6  | 20.7  | 21.0  | 19.5  | 15.7  | 12.8  | 10.6  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 5%          | DB     | 7.6   | 5.2   | 6.0   | 9.0   | 13.3  | 16.7  | 20.1  | 20.3  | 18.7  | 15.1  | 12.1  | 9.7   |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 10%         | DB     | 6.2   | 4.3   | 5.3   | 8.3   | 12.3  | 15.8  | 19.5  | 19.7  | 17.9  | 14.5  | 11.6  | 8.8   |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 5.7   | 5.2   | 4.2   | 2.8   | 2.6   | 2.7   | 2.8   | 2.5   | 2.5   | 2.8   | 3.6   | 5.2   |
|                                                                        |             | MCDBR  | 5.6   | 5.1   | 4.0   | 3.6   | 3.9   | 3.7   | 3.5   | 3.0   | 2.8   | 2.5   | 3.0   | 5.4   |
|                                                                        |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.307 | 0.313 | 0.332 | 0.357 | 0.386 | 0.429 | 0.448 | 0.416 | 0.373 | 0.343 | 0.322 | 0.302 |
|                                                                        | taud        |        | 2.463 | 2.465 | 2.462 | 2.428 | 2.393 | 2.288 | 2.255 | 2.336 | 2.429 | 2.508 | 2.532 | 2.523 |
|                                                                        | Ebn at Noon |        | 824   | 885   | 909   | 909   | 887   | 845   | 823   | 840   | 853   | 838   | 802   | 800   |
|                                                                        | Edh at Noon |        | 72    | 85    | 97    | 109   | 116   | 130   | 133   | 119   | 100   | 82    | 68    | 63    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.58  | 2.50  | 3.70  | 4.67  | 5.29  | 5.67  | 5.99  | 5.40  | 4.33  | 2.81  | 1.75  | 1.28  |
|                                                                        | RadStd      |        | 0.11  | 0.19  | 0.24  | 0.49  | 0.61  | 0.48  | 0.40  | 0.26  | 0.26  | 0.25  | 0.16  | 0.14  |

**Historical Trends**

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (39 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +72     | N/A     |

Nomenclature: See separate page